

AR 521: Control of Robotic Systems

Course Code: AR 521

Course Name: Control of Robotic Systems

L-T-P-C: 3-0-0-3

Prerequisites: For B.Tech. – Control System or equivalent course; For all – Linear Algebra

Intended for: B.Tech, M.Tech, M.Tech. (R), PhD

Mutual Exclusion: None

Approval: 48th Senate (listed in the B.Tech General Engineering and B.Tech Quantum Technology curricula); syllabus approved in 64th BoA

Preamble

Control systems play a vital role in the field of engineering, by helping regulate and manipulate a process or a system to obtain controlled outputs. Robot control systems are the brains behind robots, allowing them to perceive, plan, and execute actions in the real world. This course provides an idea about the control methodology needed specifically in the field of robotics and associated construction, through specific foundational experimental setups.

Course Objectives

- To learn foundational control theory specifically applicable to Robotic Systems
- To learn PD, PID, and State Estimation Theory for Motion Control.
- To decipher concepts related to the control of robotic manipulators.
- To understand dynamics, force control, compliance, and related concepts.

Course Modules

- **Unit 1:** Overview of Control Systems, Linear Time Invariant Systems with Single Input and Output, Feedback Control and Stability, Joint Space Control, Resolved Motion Rate Control, Natural Systems, Dissipative Systems, Passive System Stability, Trajectory generation: Determining the joint variables for desired trajectory. (8 hours)
- **Unit 2:** PD Control, Control Partitioning, Motion Control, Disturbance Rejection, Steady-State Error, PID Control, State Estimation in Feedback Systems, Effective Inertia (6 hours)
- **Unit 3:** Manipulator Control, PD Control Stability, Task-Oriented Control, Task-Oriented Equations of Motion, Operational Space Dynamics, Example, Nonlinear Dynamic Decoupling, Trajectory Tracking (8 hours)
- **Unit 4:** Compliance, Force Control, Indirect vs Direct Force Control Strategies, Dynamics, Task Description for Robot Control (12 hours)
- **Unit 5:** Control of Rotary Flexible Links and Flexible Joints, Control of Rotary Double Inverted Pendulum, Rotary Servo Base Unit for Modular Control Experiment, Gantry Crane Control System. (6 hours)

Text Books

1. Norman S. Nise, *Control Systems Engineering*, Wiley India, Student Edition (Fifth), 2009
2. *Robot Modeling and Control* by M.W. Spong, S. Hutchinson, and M. Vidyasagar, Wiley, 2006.

References

1. *Introduction to Robotics: Mechanics and Control* by J.J. Craig, Pearson, 4th Edition, 2017.
2. *Modern Robotics: Mechanics, Planning, and Control* by K.M. Lynch and F.C. Park, Cambridge University Press, 2017.